

A Test of Mean Resolution Time

Unit 4 · Topic 4.5 · TODAY'S PROBLEM

Suppose a help desk closed 1,200 tickets in a quarter. To investigate whether their population mean resolution time was below 12 hours, an analyst took an SRS of 24 and used $\alpha = 0.05$.

Leah: “The lower-tail test gives $t = -1.5747$, $df = 23$, and $p \approx 0.0645$. I would state what evidence this does and does not provide.”

Damon: “Because $p > 0.05$, accept H_0 . The test proves the population mean was exactly 12 hours and the observed difference was only chance.”

Ticket sample

Summary	Value
Population	1200
n	24
Mean (h)	11.1
SD (h)	2.8
Shape	Symmetric; no outliers

FIRST TAKE (5 min, on your sheet)

Which account is more defensible? Cite the p-value comparison or a claim that a test can or cannot justify.

- DOK 1** (a) Define the population mean and state hypotheses for the research question.
- DOK 2** (b) Verify random, 10 percent, and shape conditions. Compute SE , t , df , and the lower-tail p-value; interpret the p-value.
- DOK 3** (c) Adjudicate the accounts. Compare p with α , distinguish the formal decision from acceptance or proof, state the scoped population conclusion, and explain the two beliefs produced by the rejected account.

Video + follow-along: [u7_lesson5_live.html](https://www.khanacademy.org/a/u7_lesson5_live.html) (scan the code)

Finish (a)–(c) after the video. Turn in your sheet.

